PROBLEM K1

Stokes layer · Path lines · Fluid line · Circulation

Flow over an Oscillating Porous Wall with Transverse Blowing

Original statement · K1.pdf

OFFICIAL PROBLEM STATEMENT

A fluid is blown with velocity V through a horizontal porous wall that is oscillating in its plane with velocity $U\cos(\Omega t)$. The resulting fluid velocity for $0 < y < \infty$ is given by (approximately)

$$v_x = U e^{-y/\delta} \cos\left(\Omega t - \frac{y}{\delta}\right), \qquad v_y = V,$$

where the characteristic thickness $\delta = (2\nu/\Omega)^{1/2}$ depends on the kinematic viscosity of the fluid ν.

1. Determine the trajectory and path of a fluid particle initially located at $x = 0$ and $y = 0$.
2. Obtain the streamline that intersects the point $x = 0$ and $y = 0$.
3. Compute the vorticity $\nabla \wedge \vec{v}$. If $\nabla \wedge \vec{v} = 0$, obtain also the velocity potential.
4. Calculate the rate of expansion $\nabla \cdot \vec{v}$. If $\nabla \cdot \vec{v} = 0$, obtain also the stream function.
5. Equation of the fluid line formed by the fluid particles initially located along the vertical axis $x = 0$.
6. Determine the volume flux crossing the plane $x = 0$ as a function of time $Q(t)$ along with its average value $Q_a = \int_t^{t+2\pi/\Omega} Q\, dt$.
7. Obtain the volume flux leaving the sphere of radius R centered at $x = 0$ and $y = 2R$.
8. Compute the circulation around a square of side L whose base lies parallel to the wall at a distance $y = L/2$. Redo the computation making use of Stokes theorem.

Source: K1.pdf (page 1). Notes.pdf, Chapter 2: trajectories and path lines Eqs. (2.10)-(2.15), fluid lines Eqs. (2.16)-(2.17), stream lines Eq. (2.21), circulation and Stokes theorem Eqs. (2.29)-(2.33), potential Eq. (2.34), convective flux Eqs. (2.35)-(2.38), stream function Eqs. (2.40)-(2.45).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- The motion is planar: the velocity has components (v_x, v_y) that depend only on y and t. The only spatial degree of freedom of the field is y; the flow is unsteady with the single time scale Ω^{-1}.
- The field is given as an approximation to a viscous (Stokes-layer) solution; here it is treated as a prescribed kinematic field and no dynamics are used.
- The fluid occupies $0 < y < \infty$. At the wall, $v_x(y=0) = U\cos\Omega t$ equals the wall velocity and $v_y = V$ is the blowing velocity.
- We assume $V \neq 0$ (transverse blowing) and $\Omega > 0$, so that δ is real and positive.
- Fluxes and the circulation are computed per unit length in the direction z normal to the plane of the motion.

Symbol	Meaning	SI unit
U	amplitude of the wall velocity	m/s
V	blowing velocity (uniform, along y)	m/s
Ω	angular frequency of the wall oscillation	s^{-1}
ν	kinematic viscosity	m^2/s
$\delta = (2\nu/\Omega)^{1/2}$	thickness of the oscillatory layer	m
L, R	side of the square and radius of the sphere	m
$Q(t)$	volume flux per unit depth through $x = 0$	m^2/s
Γ	circulation	m^2/s

Dimensional check of δ: $[\nu/\Omega] = (\mathrm{m}^2/\mathrm{s})/(1/\mathrm{s}) = \mathrm{m}^2$, so δ is a length.

PHASE 2

Coordinates, Frames and Changes of Variable

- Frame of observation: inertial laboratory frame, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, with x along the wall, y normal to it and $\mathbf{e}_z = \mathbf{e}_x \wedge \mathbf{e}_y$. The wall oscillates in this frame; the velocity field given in the statement is the laboratory-frame field.
- Eulerian variables: (x, y, t). Lagrangian label of a particle: its position (x_0, y_0) at $t = 0$.
- Phase variable used throughout: $\Theta(y,t) \equiv \Omega t - y/\delta$, so that $v_x = U e^{-y/\delta} \cos\Theta$.
- Change of variable along a particle: since $v_y = V$ is constant, $y(t) = y_0 + Vt$, hence $t = (y - y_0)/V$. This trades time for y when the path line is required.
- Abbreviations: $b \equiv \Omega - V/\delta$ and $D \equiv (V/\delta)^2 + b^2$ (so $D > 0$ because $V \neq 0$).

Why a lemma is needed

Parts 1, 2, 5 and 6 require primitives of an exponential times a cosine. Instead of quoting a table, the primitive is proved once. For constants a, β, c with $a^2 + \beta^2 \neq 0$,

$$\int e^{as}\cos(\beta s + c)\,ds = \frac{e^{as}}{a^2+\beta^2}\left[a\cos(\beta s + c) + \beta\sin(\beta s + c)\right] + C. \quad \boxed{(L)}$$

Proof by differentiation: $\frac{d}{ds}\left\{e^{as}\left[a\cos(\beta s + c) + \beta\sin(\beta s + c)\right]\right\} =$
$e^{as}\left[a^2\cos + a\beta\sin - a\beta\sin + \beta^2\cos\right] = (a^2+\beta^2)\,e^{as}\cos(\beta s + c)$.

PHASE 3

Step-by-Step Derivation

3.1 Trajectory and path line of the particle starting at the origin

Why this tool: a trajectory is the solution of $d\mathbf{x}/dt = \mathbf{v}(\mathbf{x}, t)$ with $\mathbf{x}(0) = \mathbf{x}_0$ (Notes Eq. 2.12); the path line is the curve in space obtained by eliminating t (Eq. 2.15). The system is triangular: v_y does not depend on x, so $y(t)$ is found first.

Vertical motion:

$$\frac{dy}{dt} = V \implies \int_0^y dy' = \int_0^t V\,dt' \implies y(t) = Vt.$$

Along this particle, $e^{-y/\delta} = e^{-Vt/\delta}$ and $\Theta = \Omega t - Vt/\delta = bt$. Therefore

$$\frac{dx}{dt} = U e^{-Vt/\delta}\cos(bt).$$

Apply (L) with $s = t'$, $a = -V/\delta$, $\beta = b$, $c = 0$, and Barrow's rule between the limits $t' = 0$ and $t' = t$:

$$x(t) = U\left[\frac{e^{-Vt'/\delta}}{D}\left(-\frac{V}{\delta}\cos bt' + b\sin bt'\right)\right]_{t'=0}^{t'=t}.$$

Upper limit: $\dfrac{e^{-Vt/\delta}}{D}\left(-\dfrac{V}{\delta}\cos bt + b\sin bt\right)$. Lower limit: $\dfrac{1}{D}\left(-\dfrac{V}{\delta}\cdot 1 + b\cdot 0\right) = -\dfrac{V}{\delta D}$. Subtracting,

$$\boxed{x(t) = \frac{U}{D}\left[e^{-Vt/\delta}\left(-\frac{V}{\delta}\cos bt + b\sin bt\right) + \frac{V}{\delta}\right], \qquad y(t) = Vt.}$$

Path line: substitute $t = y/V$, so that $bt = (\Omega/V - 1/\delta)\,y$:

$$x(y) = \frac{U}{D}\left[e^{-y/\delta}\left(-\frac{V}{\delta}\cos\left[\left(\frac{\Omega}{V} - \frac{1}{\delta}\right)y\right] + b\sin\left[\left(\frac{\Omega}{V} - \frac{1}{\delta}\right)y\right]\right) + \frac{V}{\delta}\right].$$

The trajectory carries the time law; the path line is a damped oscillation of the horizontal displacement versus height.

3.2 Streamline through the origin

Why this tool: a streamline is tangent to $\mathbf{v}$ at a frozen instant (Notes Eq. 2.21), so t is a parameter and not an integration variable. With t frozen:

$$\frac{dx}{v_x} = \frac{dy}{v_y} \implies \frac{dx}{dy} = \frac{U}{V} e^{-y/\delta} \cos\left(\Omega t - \frac{y}{\delta}\right).$$

Apply (L) with $s = y'$, $a = -1/\delta$, $\beta = -1/\delta$, $c = \Omega t$; here $a^2 + \beta^2 = 2/\delta^2$:

$$\int e^{-y'/\delta} \cos\Theta \, dy' = \frac{e^{-y'/\delta}}{2/\delta^2} \left[-\frac{1}{\delta}\cos\Theta - \frac{1}{\delta}\sin\Theta\right] = -\frac{\delta}{2} e^{-y'/\delta} (\cos\Theta + \sin\Theta), \qquad \Theta = \Omega t - \frac{y'}{\delta}.$$

Barrow's rule between $y' = 0$ (where $x = 0$ and $\Theta = \Omega t$) and $y' = y$:

$$\boxed{x(y,t) = \frac{U\delta}{2V}\left[\cos\Omega t + \sin\Omega t - e^{-y/\delta}(\cos\Theta + \sin\Theta)\right], \quad \Theta = \Omega t - \frac{y}{\delta}.}$$

Check by differentiation, using $\partial\Theta/\partial y = -1/\delta$: $\partial_y(\cos\Theta + \sin\Theta) = (\sin\Theta - \cos\Theta)/\delta$, so $\partial_y\left[e^{-y/\delta}(\cos\Theta + \sin\Theta)\right] = -\frac{1}{\delta}e^{-y/\delta}\left[(\cos\Theta + \sin\Theta) + (\cos\Theta - \sin\Theta)\right] = -\frac{2}{\delta}e^{-y/\delta}\cos\Theta$, hence $dx/dy = (U/V)e^{-y/\delta}\cos\Theta$ as required.

3.3 Vorticity and velocity potential

Why this tool: vorticity $\boldsymbol{\omega} = \nabla \wedge \mathbf{v}$ measures twice the local angular velocity (Notes Eq. 2.52); a velocity potential $\mathbf{v} = \nabla\varphi$ (Eq. 2.34) exists only when $\boldsymbol{\omega} \equiv 0$ in a simply connected domain.

$$\omega_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = 0 - \frac{\partial v_x}{\partial y}.$$

Chain rule, using $\partial_y e^{-y/\delta} = -e^{-y/\delta}/\delta$ and $\partial_y \cos\Theta = -\sin\Theta\,\partial_y\Theta = \sin\Theta/\delta$:

$$\frac{\partial v_x}{\partial y} = U e^{-y/\delta}\left[-\frac{1}{\delta}\cos\Theta + \frac{1}{\delta}\sin\Theta\right] \implies \boxed{\omega_z = \frac{U}{\delta} e^{-y/\delta}\left(\cos\Theta - \sin\Theta\right) \neq 0.}$$

For example, at $y = 0$ and $t = 0$: $\omega_z = U/\delta \neq 0$. The flow is rotational, so no velocity potential exists.

3.4 Rate of expansion and stream function

Why this tool: $\nabla \cdot \mathbf{v}$ is the rate of change of volume per unit volume (Notes Eqs. 2.62-2.63). If it vanishes, a planar field admits a stream function with $v_x = \partial_y \psi$, $v_y = -\partial_x \psi$ (Eq. 2.41), whose level lines are stream lines (Eq. 2.42).

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 + 0 = 0.$$

Integrate $v_y = -\partial_x \psi = V$ with respect to x: $\psi = -Vx + f(y,t)$. Then $\partial_y \psi = f'(y,t) = v_x$, and by the primitive of 3.2,

$$f(y,t) = U\int e^{-y/\delta}\cos\Theta\, dy = -\frac{U\delta}{2} e^{-y/\delta}\left(\cos\Theta + \sin\Theta\right) + \psi_0(t),$$

$$\boxed{\psi(x,y,t) = -Vx - \frac{U\delta}{2} e^{-y/\delta}\left(\cos\Theta + \sin\Theta\right) + \psi_0(t).}$$

Check: $-\partial_x\psi = V$ and $\partial_y\psi = U e^{-y/\delta}\cos\Theta = v_x$. Consistency with 3.2: along the curve of 3.2, $\psi - \psi_0 = -Vx(y,t) - \frac{U\delta}{2}e^{-y/\delta}(\cos\Theta + \sin\Theta) = -\frac{U\delta}{2}(\cos\Omega t + \sin\Omega t)$, independent of y, so that curve is a level set $\psi = \text{const}$.

3.5 Fluid line initially on the axis $x = 0$

Why this tool: a fluid line is made of particles, so it is obtained by writing the trajectories (Eq. 2.17) of all particles of the initial curve $\mathbf{x}_l(\lambda) = (0, \lambda)$ and then eliminating the label λ.

Particle λ: $y = \lambda + Vt'$ and $\Theta = \Omega t' - (\lambda + Vt')/\delta = bt' - \lambda/\delta$, so

$$\frac{dx}{dt'} = U e^{-\lambda/\delta} e^{-Vt'/\delta} \cos\left(bt' - \frac{\lambda}{\delta}\right).$$

Lemma (L) with $a = -V/\delta$, $\beta = b$, $c = -\lambda/\delta$ and Barrow's rule between $t' = 0$ and $t' = t$:

$$x(\lambda, t) = \frac{U e^{-\lambda/\delta}}{D}\left\{e^{-Vt/\delta}\left[-\frac{V}{\delta}\cos\left(bt - \frac{\lambda}{\delta}\right) + b\sin\left(bt - \frac{\lambda}{\delta}\right)\right] - \left[-\frac{V}{\delta}\cos\frac{\lambda}{\delta} - b\sin\frac{\lambda}{\delta}\right]\right\},$$

where the lower limit used that cosine is even and sine is odd. Now eliminate $\lambda = y - Vt$ (the change of variable from label to current height). Then $e^{-\lambda/\delta}e^{-Vt/\delta} = e^{-y/\delta}$, $e^{-\lambda/\delta} = e^{-(y-Vt)/\delta}$ and $bt - \lambda/\delta = \Omega t - y/\delta = \Theta$:

$$\boxed{x(y,t) = -\frac{U e^{-y/\delta}}{D}\left[\frac{V}{\delta}\cos\Theta - b\sin\Theta\right] + \frac{U e^{-(y-Vt)/\delta}}{D}\left[\frac{V}{\delta}\cos\frac{y-Vt}{\delta} + b\sin\frac{y-Vt}{\delta}\right]}$$

with $\Theta = \Omega t - y/\delta$, $b = \Omega - V/\delta$, $D = (V/\delta)^2 + b^2$, valid for $y \geq Vt$ (the particles with $\lambda \geq 0$). At $t = 0$ it reduces to $x = 0$.

3.6 Volume flux through the plane $x = 0$

Why this tool: the volume flux through a fixed surface is $\int \mathbf{v}\cdot\mathbf{n}\, d\sigma$ (Eq. 2.36). The plane $x = 0$ has normal $\mathbf{e}_x$, so only v_x crosses it; per unit depth, $d\sigma = dy$.

$$Q(t) = \int_0^\infty v_x(0, y, t)\, dy = U\left[-\frac{\delta}{2}e^{-y/\delta}\left(\cos\Theta + \sin\Theta\right)\right]_{y=0}^{y=\infty}.$$

At the upper limit the exponential vanishes; at $y = 0$, $\Theta = \Omega t$:

$$\boxed{Q(t) = \frac{U\delta}{2}\left(\cos\Omega t + \sin\Omega t\right) = \frac{U\delta}{\sqrt{2}}\cos\left(\Omega t - \frac{\pi}{4}\right).}$$

Integral over one period, as written in the statement, with the primitive $\int(\cos\Omega t' + \sin\Omega t')\,dt' = (\sin\Omega t' - \cos\Omega t')/\Omega$ and Barrow's rule:

$$\int_t^{t+2\pi/\Omega} Q\,dt' = \frac{U\delta}{2\Omega}\left[\sin\Omega t' - \cos\Omega t'\right]_t^{t+2\pi/\Omega} = \frac{U\delta}{2\Omega}\left[(\sin\Omega t - \cos\Omega t) - (\sin\Omega t - \cos\Omega t)\right] = 0,$$

because $\Omega(t + 2\pi/\Omega) = \Omega t + 2\pi$. The time average (that integral divided by the period $2\pi/\Omega$) is also 0: there is no net volume transported through $x = 0$ over a cycle.

3.7 Flux leaving the sphere of radius R centred at $(0, 2R)$

Why this tool: for a closed surface the flux is turned into a volume integral by Gauss' formula (Eq. 2.37). The sphere (a body in the planar flow extended uniformly in z) has lowest point at $y = 2R - R = R > 0$, so it lies entirely inside the fluid domain $y > 0$ and the field is smooth in it.

$$\oint_\Sigma \mathbf{v}\cdot\mathbf{n}\,d\sigma = \int_{V_\Sigma} \nabla\cdot\mathbf{v}\,dV = \int_{V_\Sigma} 0\,dV = 0.$$

Independent check by symmetry: v_x depends on y only, so it is even under $x \to -x$ while n_x is odd, and $\oint v_x n_x\,d\sigma = 0$; the uniform $V\mathbf{e}_y$ gives $V\oint n_y\,d\sigma = 0$ for any closed surface. Result: $\boxed{0}$.

3.8 Circulation around the square

Why this tool: the circulation is $\Gamma = \oint \mathbf{v}\cdot d\mathbf{l}$ (Eq. 2.29); by Stokes (Eq. 2.32) it equals the flux of ω_z through the enclosed area provided the area lies inside the fluid. The square has vertices $(x_1, L/2)$, $(x_1 + L, L/2)$, $(x_1 + L, 3L/2)$, $(x_1, 3L/2)$ and is traversed counter-clockwise (normal $+\mathbf{e}_z$).

Line integral, side by side:

- bottom ($y = L/2$, $dx > 0$): $\int_{x_1}^{x_1+L} v_x\,dx = L\,v_x(L/2)$;
- right side ($x = x_1 + L$, $dy > 0$): $\int_{L/2}^{3L/2} V\,dy = VL$;
- top ($y = 3L/2$, $dx < 0$): $-L\,v_x(3L/2)$;
- left side ($dy < 0$): $-VL$.

The vertical contributions cancel, hence

$$\boxed{\Gamma = UL\left[e^{-L/2\delta}\cos\left(\Omega t - \frac{L}{2\delta}\right) - e^{-3L/2\delta}\cos\left(\Omega t - \frac{3L}{2\delta}\right)\right].}$$

Stokes theorem:

$$\Gamma = \int_{x_1}^{x_1+L}\int_{L/2}^{3L/2}\omega_z\,dy\,dx = L\int_{L/2}^{3L/2}\left(-\frac{\partial v_x}{\partial y}\right)dy = -L\left[v_x\right]_{y=L/2}^{y=3L/2} = L\left[v_x\left(\tfrac{L}{2}\right) - v_x\left(\tfrac{3L}{2}\right)\right],$$

which is the same expression.

PHASE 4

Interpretation, Limits and Dimensional Check

- Wall limit: at $y = 0$, $v_x = U\cos\Omega t$ (the wall velocity) and the oscillation decays as $e^{-y/\delta}$, with a phase lag y/δ. For $y \gg \delta$ the fluid has no motion in x and only the uniform blowing V remains.
- Limit $V \to 0$ in 3.1: $D \to \Omega^2$, $b \to \Omega$ and $x(t) \to (U/\Omega)\sin\Omega t$, the displacement of the wall itself, as expected for a particle that stays on the wall.
- Limit $V \to 0$ in 3.2: $x \propto 1/V \to \infty$, i.e. the streamlines become parallel to the wall, consistent with $v_y \to 0$.
- The flow is incompressible and rotational; it has a stream function but no potential, as found in 3.3 and 3.4. The zero net flux in 3.7 and the zero cycle integral in 3.6 follow from this solenoidal, periodic character.
- Dimensions (SI): $[x(t)] = [U/D]\,[V/\delta] = (\mathrm{m/s})(\mathrm{s^2})(\mathrm{s^{-1}}) = \mathrm{m}$. $[\omega_z] = [U/\delta] = \mathrm{s^{-1}}$. $[\psi] = [Vx] = [U\delta] = \mathrm{m^2/s}$. $[Q] = [U\delta] = \mathrm{m^2/s}$ per unit depth. $[\Gamma] = [UL] = \mathrm{m^2/s}$.
- Numerical cross-check (quadrature of the defining integral against the closed form), with $U = 1.3$ m/s, $V = 0.7$ m/s, $\delta = 0.9$ m, $\Omega = 2.1$ s^{-1}, $t = 1.7$ s: path $x(t) = 0.653175$ m (both); streamline at $y = 0.8$ m, $x = -0.952255$ m (both); fluid line (valid only for $y \geq Vt = 1.19$ m) at $y = 1.5$ m, $x = 0.618267$ m, and at $y = 2.5$ m, $x = 0.194357$ m (both); $Q = -0.775155$ m^2/s (both); for $L = 0.6$ m, $\Gamma = -0.315037$ m^2/s from the line integral, from the closed form and from the Stokes integral.

Reading of the official handwritten solution

In K1.pdf the first bracket of part 5 is written $\frac{V}{\delta}\cos[\cdot] - (\Omega - \frac{V}{\delta})\sin[\cdot]$ and, as legible in the scan, the argument of that sine shows y where t is meant. The expression above is the result of the explicit Barrow derivation. The statement of part 6 defines Q_a as the bare integral over one period (no factor $\Omega/2\pi$); the handwritten answer is 0, which agrees with the integral computed above, and the mean value is also 0.